

Cladding

Cladding means a facing or architectural decoration additional to the structural elements of a building, e.g. metal cladding, stone cladding, glass-fibre reinforced concrete (GRC) cladding and the like. Cladding should comply with the performance requirements stipulated in sections 3 and 28 of the Building (Construction) Regulation in respect of materials, fixings, strength, durability, non-combustibility, means of access for maintenance or repair, etc.

2. Cladding should be provided with sufficient permanent flexible horizontal and vertical joints to accommodate differential movement between the cladding and the structure of the building to which it is attached. All anchors, dowels and fixings should be of stainless steel or other corrosion resistant materials. They should be suitable, permanent and adequately protected from corrosion. Requirements on testing of anchors and cladding panels are given in paragraphs 10 to 16 below.

Design and Construction Standards

3. Standards commonly used for the design and construction of cladding works, which are acceptable to the Building Authority (BA), are given in **Appendix A**. Regarding non-combustibility, the standards acceptable to the BA are given in section 5 of Part E of the Code of Practice for Fire Safety in Buildings 2011 (FS Code). Clauses E10.1 and E12.1 of the FS Code are applicable to external cladding while clause E13.1 of the same sets out the deem-to-satisfy provisions for internal cladding.

Submission of Cladding Plans

4. Where cladding is to be affixed to any part of the exterior of a building, details such as the location, layout, elevation and material should be shown in the general building plan (GBP) submitted to the BA for approval. When the cladding to be installed is above 6 metres from the adjoining ground level or adjoining floor, in addition to the building plans, details such as the thickness, strength, durability, and type of the cladding, material of fixings and layout of the support to the cladding should be shown in the structural plans submitted for approval. Failure to do so may result in delay in or refusal of approval/consent of the cladding plans. As regards the fixings of stone cladding, sand/cement bedding and/or epoxy bonding alone is not considered a suitable and permanent fixing.

5. The following details are required to be included in the structural plans of cladding for submission to the BA for approval:

- (a) structural framing and details of structural elements *excluding any unnecessary shop fabrication details*;

/(b) ...

- (b) elevations including location of expansion joints;
- (c) anchors and support details;
- (d) design loads, design standards and codes of practice;
- (e) material specifications for metal (e.g. structural steel, aluminium alloy), stone (e.g. granite/marble/limestone), GRC and anchors;
- (f) workmanship specifications for welding, corrosion protection (e.g. galvanisation) and measures to overcome bi-metallic effects; and
- (g) structural calculations comprising design check on the parent structure, analysis on the structural adequacy and stability of the proposed cladding system, element design for metal, stone or GRC cladding and deflection check on major load carrying members, if applicable.

Streamlined Plan Processing of Metal Cladding

6. To streamline the plan processing of metal cladding, registered structural engineers (RSE) may include the general notes, typical layout and typical details of metal cladding in the superstructure plans instead of making a separate submission of structural plans of metal cladding for approval. While the location, layout and elevation of metal cladding should be shown on the GBP for approval, the details as mentioned in paragraph 5 above should be included in the superstructure plans. The authorized person (AP) and RSE of the project should ensure that the works so approved are compatible with the approved GBP and do not contravene any provisions of the Buildings Ordinance (BO) and its subsidiary regulations. Reference can also be made to the checklist at Appendix B6 and the sample drawing at Appendix B9 of Practice Note for Authorized Persons, Registered Structural Engineers and Registered Geotechnical Engineers (PNAP) ADV-33.

Separate RSE

7. In view of the speciality of cladding works, a separate RSE may be appointed to prepare the design and to supervise the carrying out of such works. Under such circumstances, Forms BA4 and BA5 indicating the appointment of the separate RSE and the scope of works for which he/she is responsible are required to be submitted together with the plans submitted for approval. Upon completion of the cladding works, the separate RSE so appointed is required to certify satisfactory completion of the works in accordance with regulation 25(3) of the Building (Administration) Regulations.

8. When structural details for cladding works are submitted by the separate RSE, the submission should be accompanied by an assessment report prepared by him/her giving an account of the effect of the proposed works to the parent structure. The assessment report should be appended with a statement signed by the project RSE to confirm that he/she is fully aware of the connection details and the effect of the works on the parent structure.

9. The separate RSE who is appointed for the cladding works should be responsible for the supervision of the construction of such works, including the installation of any cast-in anchorage, e.g. anchor plates, cast-in embeds, through bolts, etc., in the parent structure except in the event where the cast-in anchorage has been pre-installed in the parent structure prior to his/her appointment. For such cases, the structural details and layout of the pre-installed parts should be given in the superstructure plans to be submitted by the project RSE to the BA for approval. The project RSE should then be responsible for the supervision of the installation of such parts. The separate RSE should refer to the pre-installed connection details when designing the cladding works and should co-ordinate with the project RSE for necessary amendment if different connection details are to be used.

Tests on Anchors and Cladding Panels

Anchors

10. On-site strength tests should be carried out on a representative number of each type and size of drilled-in anchors for those parts of the cladding to be installed above 6 metres from the adjoining ground level or adjoining floor. Such tests are necessary for verifying the performance and workmanship of the anchors installed and should be carried out under the direction of the RSE or AP.

11. Each representative anchor should be tested by pull-out test to demonstrate that its pull-out capacity is not less than 1.5 times the recommended tensile load as specified by the anchor manufacturer. The tested anchor should be considered satisfactory if it does not show any signs of separation, plastic deformation or deleterious effect during the test.

Stone Cladding Panels

12. Stone is a natural material. The mechanical properties, physical properties and chemical properties can vary considerably between different types and grades of stones. Tests on stone cladding panels are required to be carried out to verify the average and characteristic strengths adopted in the design and to form part of the quality assurance and control during construction. The RSE and registered contractor (RC) responsible for the cladding works should appoint a grade T3 technically competent person (TCP) under the RSE's stream and a grade T1 TCP under the RC's stream respectively on site to detect any deleterious or potentially unstable constituents such as clay-like minerals, iron minerals or veins on the stone panels delivered to site.

13. When stone cladding is to be used, the following tests are required to be carried out for each type of stone, the test results of which should be checked against the acceptance criteria given in **Appendix B**:

- (a) flexural strength test of dimension stone to ASTM C880 – Standard Test Method for Flexural Strength of Dimension Stone or to BS EN12372 – Natural Stone Test Methods, Determination of Flexural Strength under Concentrated Load or to other appropriate equivalent standards; and

/(b) ...

- (b) strength test of individual stone anchorage to ASTM C1354 – Standard Test Method for Strength of Individual Stone Anchorages in Dimension Stone or to other appropriate equivalent standards.

14. As limestone is highly susceptible to acid rain and may deteriorate quickly when being used as exterior cladding material, in addition to the tests required in paragraph 13 above, aged strength testing to simulate weathering due to extreme temperature change and extreme moisture content change should be carried out for limestone cladding to demonstrate that the residual flexural and anchorage capacities are not less than 80% of those obtained in the standard tests specified in paragraph 13 above. The aim of aged strength testing is to simulate the conditions in which limestone panels are fully saturated and return to their driest conditions and to simulate the critical temperature changes during their intended life. Details of the aged strength test are given in **Appendix B**.

15. At least 5 test specimens should be selected at random from batches of stone delivered to site for each of the test specified in paragraphs 13(a), 13(b) and 14 above.

16. The test should be carried out by or under the direction and supervision of a testing agency independent of the supplier of the stone cladding. The test results should be certified by the testing agency, and endorsed by the AP/RSE to confirm that the test results have reached the required characteristic strengths adopted in the design. The test reports should be submitted to the BA prior to the application for occupation permit.

Minor Works Relating to External Cladding

17. Under the Minor Works Control System, certain minor building works relating to erection, repair, replacement or removal of cladding are designated as minor works, which may be carried out under the simplified requirements as an alternative to obtaining prior approval and consent under the BO. Reference can be made to Schedule 1 of the Building (Minor Works) Regulation and PNAP APP-147 regarding the list of minor works items and the simplified requirements respectively.

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**Standards Commonly Used for the
Design and Construction of Cladding Works
Acceptable to the Building Authority**

Reference Standards

The standards listed in this appendix are intended to provide reference information only for the purpose of design and construction of cladding works. It should be noted that:

- (a) the standards listed are not meant to be exhaustive;
- (b) national standards and codes of practice of various countries, though similar in major aspects, do not have exact equivalence to one another;
- (c) should a certain design standard be adopted, it should be applied to the design consistently; and
- (d) Buildings Ordinance and its subsidiary regulations always take precedence over other design standards should there be a conflict between them.

A. Material and Workmanship

Material and Workmanship	Standard	Title
Steel	Hong Kong Code of Practice	Code of Practice for the Structural Use of Steel 2011
Aluminium	BS 1161:1977(1991)	Specification for aluminium alloy sections for structural purposes
	BS EN 515:2017, BS EN 485-3:2003, BS EN 573-4: 2004, BS EN 573-2:1995, BS EN 485-2: 2016+A1:2018	Specification for wrought aluminium and aluminium alloys for general engineering purposes: plate, sheet and strip
	BS 1473:1972(2002)	Specification for wrought aluminium and aluminium alloys for general engineering purposes – rivet, bolt and screw stock

Material and Workmanship	Standard	Title
Aluminium	BS EN 573-4:2004, BS EN 573-3:2019, BS EN 755-6:2008, BS EN 755-3:2008, BS EN 755-4:2008, BS EN 755-2:2016, BS EN 755-8:2016, BS EN 755-7:2016, BS EN 12020-2:2016, BS EN 755-9:2016	Specification for wrought aluminium and aluminium alloys for general engineering purposes: bars, extruded round tubes and sections
	BS EN 1011-4:2000	Specification for MIG welding of aluminium and aluminium alloys
	BS EN 1999-1-1:2007+A2:2013	Structural use of aluminium – Specification for materials, workmanship and protection
Stainless Steel	BS EN 10259:1997, BS EN 10095:1999, BS EN 10048:1997, BS EN 10258:1997, BS EN 10029:2010, BS EN 10051:2010	Specification for stainless and heat-resisting steel plate, sheet and strip
	BS 6105:2016	Specification for corrosion-resistant stainless steel fasteners
	BS EN 1011-3:2018	Specification for fusion welding of austenitic stainless steels
Sealant	BS 6213: 2000+A1:2010	Selection of construction sealants. Guide

B. Design and Reference

Design	Standard/Reference	Title
Steel	Hong Kong Code of Practice	Code of Practice for the Structural Use of Steel 2011
Aluminium	BS 8118-1:1991	Structural use of aluminium – Code of practice for design

Design	Standard/Reference	Title
Natural Stone	BS 8298-4:2010, BS 8298-3:2010, BS 8298-2:2010, BS 8298-1:2010	Code of practice for design and installation of natural stone cladding & lining
	Reference	“Guide to the Selection & Testing of Stone Panels for External Use” published by Center for Window and Cladding Technology, University of Bath (1997)
Design for Safety	Hong Kong Code of Practice	Code of Practice on Access for External Maintenance 2021

C. Testing

Testing	Standard	Title
Sealant	BS 3712	Building and construction sealants
	Part 1:1991(1996)	Part 1 : Methods of test for homogeneity, relative density and penetration
	Part 2:1973(2000)	Part 2 : Methods of test for seepage, staining, shrinkage, shelf life and paint ability
	Part 3:1974(2000)	Part 3 : Methods of test for application life, skinning properties and tack-free time
	Part 4:1991(1996)	Part 4 : Method of test for adhesion in peel
	BS 5080	Structural fixings in concrete and masonry
	Part 1:1993	Part 1 : Method of test for tensile loading
	Part 2:1986(1993)	Part 2 : Method for determination of resistance to loading in shear

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Stone Cladding Panels

Acceptance Criteria for Tests on Stone Cladding Panels

1. The tests carried out in accordance with paragraph 12 of this practice note are considered acceptable if the test results comply with the following requirements:

- (a) the characteristic flexural strength is greater than 3 times the design allowable flexural strength;
- (b) the average flexural strength is greater than the design allowable flexural strength multiplied by the flexural safety factor (FSF);
- (c) the characteristic anchorage strength is greater than 4.2 times the design allowable anchorage strength; and
- (d) the average anchorage strength is greater than the design allowable anchorage strength multiplied by the anchorage safety factor (ASF).

2. In respect of paragraph 1 above, the following applies:

- (a) Characteristic strength = Average strength – $K \times \sigma$

where

K is the K-factor for at least 5 test specimens and may be taken as 3.41 (K-factor corresponding to larger numbers of test specimens may be used if appropriate); and

σ is the standard deviation.

- (b) FSF and ASF may be obtained as follows:

FSF = variation factor (VF) $\times$ durability factor (DF)

ASF = FSF $\times$ 1.4

where VF and DF are obtained from the tables below:

Coefficient of variance	VF		
	Granite	Limestone	Marble
0% - 5%	2.0	3.0	2.5
5% - 10%	2.5	3.5	3.0
10% - 20%	3.0	4.0	3.5
above 20%	3.5	4.5	4.0

Fraction of Initial Flexural Strength	DF
100%	1.0
95% - 75%	1.2
75% - 60%	1.5
Less than 60%	1.8

Where substantiation is not available, the coefficient of variance should be taken as 10% - 20%, and the fraction of initial flexural strength should be taken as (a) less than 60% for limestone and marble, and (b) 75% - 60% for granite. Accordingly, the FSF and ASF evaluated for various types of stone panels are summarised below:

	FSF	ASF
Granite	$3.0 \times 1.5 = 4.5$	$4.5 \times 1.4 = 6.3$
Limestone	$4.0 \times 1.8 = 7.2$	$7.2 \times 1.4 = 10.08$
Marble	$3.5 \times 1.8 = 6.3$	$6.3 \times 1.4 = 8.82$

Procedures for Aged Strength Tests for Limestone Panels

3. The following procedures for aged strength tests for limestone panels may be adopted on the condition that the fraction of initial flexural strength for assessment of FSF and ASF shall be taken as less than 60%:

Artificial ageing of 50 thermal cycles

- (a) Samples to be tested will be dried for 4 hours at 77°C ($\pm 2^\circ\text{C}$). The samples will be removed from the oven and cooled for 4 hours at 0°C ($\pm 2^\circ\text{C}$) in a chamber. Repeat the procedures for 50 cycles, then inspect and record the samples for any sign of damage.

Artificial ageing of 50 wet/dry cycles

- (b) Samples to be tested will be immersed in water for 4 hours at 23°C ($\pm 2^\circ\text{C}$). The samples will be removed from the water tank and placed under forced air drying (or drying by oven) for 4 hours at 23°C ($\pm 2^\circ\text{C}$). Repeat the procedures for 50 cycles, and then inspect and record the samples for any sign of damage.

Carrying out strength testing

- (c) Strength testing of the samples should be carried out after completion of the artificial ageing of 50 thermal cycles and 50 wet/dry cycles to demonstrate that the residual strengths comply with the requirements specified in paragraph 13 of this practice note.

4. Alternatively, the RSE responsible for the cladding works may propose testing procedures for acceptance by the BA prior to the commencement of the test.

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